CS 381 Project 3 **Fall 2026**

Due: Nov 3, 2026

Some test cases:

1. board: [[1,0,0,0],[2,0,0,0],[0,2,1,0],[0,0,0,0]] is a win for player 1 with best move (1,1). Can be tested with minmax.

2. board: [[0,0,0,0], [[0,0,0,0], [0,0,0,0], [[0,0,0,0]] is a win for player 1. minmax may take too long. But alphabeta search with depth 12 reveals a win for Player 1 with the move (0, 0). At depth 8 and 10, alphabeta returns a root value of 0.

3. board: [[0,1,0,0],[0,0,2,0],[0,0,0,0],[0,1,0,2]] is a draw (This can be deduced by running alphabeta pruning to depth 12. At depth 10, alphabeta pruing returns a minmax value of 0.)

4. board: [[2,1,2,1],[1,0,0,0],[2,0,0,0],[1,0,0,0]] is a win for player 1. (Can be deduced by running minmax.)

5. board [[1,2,1,0],[2,0,0,0],[0,0,0,0],[0,0,1,0]] is a win for player 2. (Can be deduced by running minmax.

At depth 6, alphabeta returns a minmax score = 4.